

Ashraful Islam

AI Engineer | Machine Learning & DataOps Intern | Generative AI & Agentic Systems

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SUMMARY

Aspiring **AI Engineer** and CSE student at CUET with research publications in **RAG, Agentic AI, and Computer Vision**, and industry experience in **Machine Learning & DataOps at Poridhi.io**. Proficient in **Python, Generative AI, Agentic AI (LangChain, LangGraph)**, **LLM Fine-Tuning**, and **MLOps (Docker, FastAPI, CI/CD)**. Proven track record of developing scalable data pipelines and deploying production-ready AI services.

WORK EXPERIENCE

Poridhi.io

Machine Learning & DataOps Intern

Dhaka, Bangladesh (Hybrid)

August 2026 – Present

- Engineered automated **DataOps pipelines** for data ingestion, quality validation, and preprocessing for ML workflows.
- Developed and containerized intelligent AI microservices and model inference REST APIs using **Python, FastAPI**, and **Docker**.
- Implemented **MLOps** workflows, including experiment tracking, containerized environments, and automated CI/CD deployment.
- Researched and prototyped **Agentic AI** and **LLM** workflows (LangChain, LangGraph) for internal automation and developer tooling.

EDUCATION

Chittagong University of Engineering and Technology (CUET)

B.Sc. in Computer Science and Engineering – CGPA: 3.52 / 4.00 (122 Credits Completed)

2023 – Present

Chittagong, Bangladesh

Undergraduate Thesis: BondhuAI-BN: Bengali Mental Health Counseling via Scoped Multi-Agent RAG with LoRA Fine-Tuning.

Dhaka City College

Higher Secondary Certificate (HSC) – Science – GPA: 5.00 / 5.00

2019 – 2021

Dhaka, Bangladesh

TECHNICAL SKILLS

- Programming Languages:** Python, C++, C, JavaScript, SQL
- Generative AI & LLMs:** Agentic AI, LangChain, LangGraph, Multi-Agent Systems, RAG, Prompt Engineering, LLM Fine-Tuning (LoRA), Vector DBs (FAISS, Chroma), Hugging Face, Gemini API
- Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, Computer Vision (Vision Transformers/DeiT), NLP, Reinforcement Learning (DQN), Explainable AI (SHAP)
- MLOps & DataOps:** Docker, FastAPI, MLflow, CI/CD Pipelines, Data Engineering, Optuna, Model Deployment, Linux, Git/GitHub
- Backend & Databases:** Django, Django REST Framework, PostgreSQL, MySQL, SQLite, MongoDB, RESTful APIs, React

FEATURED AI / ML PROJECTS

BondhuAI-BN: Scoped Multi-Agent RAG System

🐙 GitHub

Python, LangChain, LangGraph, LoRA Fine-Tuning, Hugging Face, PyTorch, FastAPI

- Architected a culturally-aware Bengali mental health support system using an **Agentic AI** multi-agent orchestration framework for intent routing, empathetic response synthesis, and clinical safety guardrails.
- Implemented domain-adaptive fine-tuning (LoRA) on open LLMs with scoped vector retrieval to eliminate hallucinations.

CCMK-RAG: Content-Conditioned Watermarked RAG

🐙 GitHub

Python, PyTorch, Transformers, FAISS, LangChain, RAG Pipelines

- Developed a passage-level attribution scheme using content-conditioned multi-key watermarks for source provenance in LLM answers.
- Demonstrated state-of-the-art attribution precision across legal corpora; accepted at **IEEE OMLET 2026**.

Personal AI Assistant: Agentic Workflows & Automation

🐙 GitHub

Gemini API, LangChain, Streamlit, OOP Architecture, Vector Search, Tool Calling

- Built an interactive conversational agent featuring multi-turn memory, tool execution, and dynamic context grounding via Streamlit.

Intelligent Traffic Light Control via Deep Reinforcement Learning

🐙 GitHub

Python, Deep Q-Networks (DQN), SUMO Simulator, PyTorch/TensorFlow, NumPy

- Implemented adaptive traffic signal optimization using DQN in SUMO simulator, reducing vehicle wait times by over 25%.

SELECTED PUBLICATIONS, CERTIFICATIONS & LEADERSHIP

- Accepted IEEE Papers:** CCMK-RAG: Watermarking for Attribution in RAG (IEEE OMLET 2026) | Section-Aware RAG for Labour Law QA (IEEE OMLET 2026) | Cyber Attack Detection in IoT Networks (IEEE ICDSBS 2026) | Counterfeit Banknote Detection via DeiT-Tiny (ICCPCT 2026).
- Certifications:** Full Stack Data Science with GenAI (Inception BD) | Complete Data Science, ML, DL, NLP Bootcamp (Udemy) | SQL Intermediate & Basic (HackerRank) | Database Structures with MySQL (Meta).
- Competitive Programming & Leadership:** Solved **500+** problems across Codeforces (*The_phoenix*, rating: 1133), LeetCode, and HackerRank | **General Secretary**, Cox's Bazar Student Forum, CUET.